

Problem 11.3

Find the accumulated value of 1 at the end of 19 years if the force of interest is

$$\delta_t = 0.04(1+t)^{-2}.$$

Solution

The accumulation function satisfies

$$\ln a(t) = \int_0^t \delta_x dx.$$

Therefore,

$$\begin{aligned}\ln a(t) &= \int_0^t 0.04(1+x)^{-2} dx \\ &= -0.04(1+x)^{-1} \Big|_0^t \\ &= -\frac{0.04}{1+t} + 0.04.\end{aligned}$$

Exponentiating both sides gives

$$a(t) = \exp \left(0.04 - \frac{0.04}{1+t} \right).$$

At the end of 19 years,

$$\begin{aligned}a(19) &= \exp \left(0.04 - \frac{0.04}{20} \right) \\ &= e^{0.038} \\ &\approx 1.038731233.\end{aligned}$$

Hence,

$$\boxed{a(19) \approx 1.038731233}.$$